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Phoebe Friesen, Lisa Kearns, Barbara Redman & Arthur L. Caplan

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Rethinking the Belmont Report?

Phoebe Friesen, NYU School of Medicine

Lisa Kearns, NYU School of Medicine

Barbara Redman, NYU School of Medicine

Arthur L. Caplan, NYU School of Medicine

This article reflects on the relevance and applicability of the Belmont Report nearly four decades after its original publication. In an exploration of criticisms that have been raised in response to the report and of significant changes that have occurred within the context of biomedical research, five primary themes arise. These themes include the increasingly vague boundary between research and practice, unique harms to communities that are not addressed by the principle of respect for persons, and how growing complexity and commodification in research have shed light on the importance of transparency. The repercussions of Belmont's emphasis on the protection of vulnerable populations is also explored, as is the relationship between the report's ethical principles and their applications. It is concluded that while the Belmont Report was an impressive response to the ethical issues of its day, the field of research ethics involving human subjects may have outgrown it.

Keywords: research ethics, Belmont Report, autonomy, beneficence, justice, human subjects research

As the new revisions to the Common Rule take their place, it seems timely to reflect on the significant document that gave shape to the regulations both at its inception and now. Since its release in 1978 and publication in 1979, the Belmont Report (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research 1979) has had an enormous impact on the way research with human subjects has been conducted in the United States and in many other nations. This article explores some of the more significant criticisms that have been raised in response to the report, as well as changes that have reshaped many questions central to research ethics since its publication, and asks whether the report still holds up nearly four decades later. Part I briefly summarizes the central components of the Belmont Report, describes the ethical climate at the time, and provides clarification about what the report was and was not meant to achieve. Part II explores several criticisms that have been leveled against the report and situates many of them within the dynamic landscape of biomedical research. Finally, Part III offers a discussion of whether the Belmont Report is in need of a 21st-century overhaul.

PART I: THE BELMONT REPORT

The 11 members of the National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research who authored the Belmont Report

worked on it from 1974 to 1978, including four intensive working days at the Smithsonian Institution's Belmont Conference Center in 1976. The Secretary of Health, Education, and Welfare (now Health and Human Services [HHS]) gave them four primary tasks: identify the boundary between research and practice, determine the role of risk-benefit analyses in human subjects research, outline appropriate guidelines for subject selection, and provide criteria for what constitutes truly informed consent (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research 1979). Both the organization and the content of the Belmont Report reflect these directives. The first section of the report addresses the first task; it defines "practice" as an intervention solely meant to enhance a patient's well-being, and "research" as "an activity designed to test an hypothesis, permit conclusions to be drawn, and thereby to develop or contribute to generalizable knowledge" (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research 1979). The second section describes three ethical principles that are "particularly relevant" to research involving human subjects: respect for persons, beneficence, and justice (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research 1979). The third links each of these principles to the remaining three tasks, suggesting that respect for persons should be the guiding principle behind informed consent, that beneficence should underlie

risk–benefit analyses, and that justice ought to be the central principle behind subject selection.

To understand the Belmont Report, one must understand the ethical climate in which it was written. Three decades earlier, the Nuremberg Code, a set of research ethics principles to guide human experimentation, had been promulgated, at the conclusion of the “Doctors Trial” of Karl Brandt and 22 other Nazi defendants (The Nuremberg Code 1949). The verdict, with some modifications, was later endorsed by the World Medical Association as a code of ethics meant to guide research involving human subjects: the Declaration of Helsinki (WMA General Assembly 1996). While there is certainly ethical overlap between these documents and the Belmont Report, the ethical emphases of the latter are drawn from particular concerns that were paramount in the United States at the time. Especially important were the revelations contained in the now famous exposé on subject abuses penned by Henry Beecher in the *New England Journal of Medicine* and the publicity surrounding the Tuskegee Syphilis Experiment, which continued to track the progression of untreated syphilis in hundreds of poor black men long after a cure had become widely available (Beecher 1966). The public outcry in 1972, triggered by an African American whistleblower at the Centers for Disease Control (McCallum et al. 2006), led Congress to establish the National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research.

The content of the Belmont Report reflects the committee’s fears that research subjects would continue to be deceived, harmed, or otherwise exploited by investigators, a point that has often been overlooked by critics. As one Belmont author explains, the Belmont Report was meant to be “a proclamation that had to ring true in the ears of scientists, policymakers, politicians, ethicists, journalists, and judges” (Jonsen 2005). As well as being a proclamation, the Belmont Report was primarily intended to be a statement of a general, principled moral framework that would help prevent abuses such as Tuskegee from occurring in research involving human subjects in the future (Beauchamp 2008). Although some have expressed concern that the report is “not easily understood or fathomed” (Vanderpool 2001), and that its principles are often misinterpreted as direct action guidelines (Levine 2005), the authors maintain that there “was never any ambition or attempt to make this document specific and practical” (Beauchamp 2005) and that the report “was not to be a philosophical treatise nor was it to be a set of guidelines” (Jonsen 2005). The Belmont Report was not written as an extension of any particular moral theory (Levine 2005), and was both principlist and casuist at the same time (Beauchamp 2005).¹ This may seem somewhat unambitious given the stakes involved, but the goal of the report was only to offer a preliminary, protective ethical

framework. A national Ethics Advisory Board was meant to act as a standing agency to continue to grapple with practical ethical issues related to human subjects research (Jonsen 2005; National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research 1978, January 13), and indeed many remaining issues were explored in 16 volumes the National Commission later produced (Beauchamp 2008). Despite its modest goal, the Belmont Report was an impressive response to the ethical issues facing human subject researchers when considered in the very troubling context in which it was written.

PART II: CRITICISMS AND CHANGES

Several significant changes have occurred within the field of human subjects research since the publication of the Belmont Report, and many concerns about the adequacy of the report have been raised in the literature in the intervening years. Five key themes that arise are the distinction between research and practice, harms to communities, the importance of transparency, implications of protectionism, and the relationship between Belmont’s ethical principles and their applications.

Research and Practice

Although the commission knew it could not adequately delineate the complicated boundary between research and practice, it did manage to capture paradigmatic features of each. The report states that research is meant to contribute to generalizable knowledge, while practice is meant to contribute to patient well-being. The authors recognized that there often could be overlap between the two; for example, practice could contain significant risks and require oversight, or could involve goals beyond the well-being of individual patients (Beauchamp and Saghai 2012). However, in the decades since the publication of the report, the line between research and practice has become even more vague.

Institutional review boards (IRBs), charged with the protection of human research subjects, have long relied on the report’s distinction to determine what is and is not within their purview, but today face an increasing number of examples of novel “practice” involving significant risks and therefore in need of significant oversight (e.g., pre-approval, or “compassionate” access to drugs; innovative treatment design; innovative surgeries), as well as a growing number of cases of “research” involving only minimal risk (e.g., retrospective analysis of deidentified data, some forms of survey research, the use of biospecimens) (Faden et al. 2013). These examples illuminate the difficulties in drawing a boundary between research and practice on the basis of the aim of either creating generalizable knowledge or seeking patient well-being. This boundary also limits what information is gained, since practitioners often abstain from collecting data during routine care or innovative practice as a result of the perceived burdens of research oversight (Rhodes 2010). As a result, risky interventions take place with no oversight and no data

1. One of the authors of the report, A. R. Jonsen, has noted that the Belmont Report was written as if guided primarily by principles, but constructed on the basis of casuistry (Jonsen 1988.)

collection, while minimal-risk research is delayed or forgone due to the difficulty of securing regulatory approval. In response, some have argued that unique guidelines for innovative practice are needed (Taylor 2010), especially since disadvantaged groups are particularly at risk of being enrolled in such therapies (Sherwin 2005). Others suggest that research and practice should be integrated into a single, comprehensive learning health care system (Faden et al. 2013).

Harms to Communities

Another criticism of the Belmont Report is that it fails to account for the unique harms to communities that can occur due to research (Weijer 1999; Weijer and Anderson 2002; Emanuel and Weijer 2005). These are not simply the aggregate of harms to individuals within a community, but harms suffered by a community as a whole. Many communities have endured a long history of abuses at the hands of researchers, leading them to doubt the likelihood that they will be beneficiaries of research conducted on them, while others do not share the goals of the researchers. These groups include poor communities, racial and ethnic minorities, those living with mental or physical illness or disabilities, and other stigmatized groups. Harm to these groups cannot be mitigated simply by securing informed consent from individuals according to Belmont's principle of respect for persons. This principle fails to take into account the unique harms that can be done to communities *qua* communities, such as violating widespread trust or taking ownership of a community's stories. For example, in the case of the controversial Human Genome Diversity Project, several indigenous communities expressed reluctance to help researchers in their development of a theory that may offer them no benefit and may serve as evidence to debunk their beliefs about their creation beliefs and histories (Weijer and Anderson 2002). This has led some to argue that there are "special issues related to the religious beliefs, cultural traditions, and history of aboriginal communities" that are not addressed by the report (Weijer and Anderson 2002). Others suggest adding a principle of respect for cultures (Levine 1982) or expanding respect for persons to "respect for persons and communities" (Lebacqz 2005).

Transparency in Research

Concerns about transparency in research have become paramount in the years since the publication of the Belmont Report. Protocols submitted to IRBs have become increasingly elaborate, while biomedicine has become much more commodified, raising issues of conflicts of interest.

When the report was written, most submissions to IRBs were for government-funded, National Institutes of Health (NIH) projects taking place at a single site. Now committees often review complex protocols that are privately sponsored and global in scope (Emanuel 2015). Further issues complicating transparency are worries about the composition of IRBs and their scope of responsibilities

as additional committees devoted to research oversight have been created, such as data safety and monitoring boards (DSMBs) and institutional biosafety committees (IBCs). Biomedicine has also seen tremendous growth in the value of patents, devices, and stock.

The majority of biomedical research in the United States is funded by industry, tying the profit motive to every stage of the research process (Taylor 2010), and making transparency much more difficult to attain. This has made the task of identifying and managing conflicts of interest increasingly important (Dumit 2012). Some have argued that the principles offered by the Belmont Report are unable to uphold research integrity, since they do not provide "a clear basis for investigators to keep promises, tell the truth, and avoid killing" (Veatch 2005).

Another significant development that compromises transparency has been the rise of new technologies that pose novel and sometimes unpredictable risks to research participants' privacy and confidentiality of data. These risks can arise throughout all stages of research: participant recruitment through social media, the development of digital monitoring devices that collect data on unwitting bystanders, the use of vast amounts of data being collected and analyzed by companies without individual consent, and the potential to reidentify data using the wealth of personal information available online or through the use of genetic data (Kelly et al. 2013; Gelinis et al. 2017; Benitez and Malin 2010; Metcalf and Crawford 2016).

Critics have noted the Belmont Report's limitations with regards to transparency, suggesting that the report pays little attention to the importance of trust and openness within research (Faden, Mastroianni, and Kahn 2005). Many of these issues, unheard of in the 1970s, threaten the ethical principles espoused in the report, especially that of beneficence. Obstacles to transparency may limit the ability of members of IRBs, DSMBs, and IBCs to adequately analyze the risks and benefits involved in a given study, if they lack access to a significant portion of past research, as a result of publication biases or trade secrets, or if they are unaware of all known risks to participants, as a result of a lack of full disclosure by private sponsors.

Implications of Protectionism

Given the Tuskegee Syphilis Experiment's role in the creation of the Belmont Report, it is understandable that the protection of vulnerable populations took center stage in the report, including those who are vulnerable due to a lack of autonomy, such as children and individuals who are mentally disabled, and those vulnerable for other reasons, such as "racial minorities, the economically disadvantaged, the very sick, and the institutionalized" (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research 1979). Certainly, protecting vulnerable research participants from exploitation or abuse is an essential task of any research ethics framework. In retrospect, however, the weight the report places on protectionism has both helped

and hindered epistemic and ethical progress in research. One negative result of the emphasis on protectionism is that some groups considered vulnerable have been left out of the research process and health data relevant to them are now lacking (e.g., pregnant women, children, embryos, the institutionalized, the elderly) (Beauchamp 2008; Rhodes 2010; Sherwin 2005; King 2005). Additionally, as the AIDS epidemic showed, being protected from research is not always desired by members of vulnerable groups (Beauchamp 2008; King 2005; Rogers and Lange 2013).

Today, the desire in many communities to move away from protectionism can be seen in the increasing demands for pre-approval access to investigational drugs, in participants' insistence that they receive active agents and not placebos in clinical trials, and in concerns about the underrepresentation of minorities in late-stage clinical research (Rogers and Lange 2013). This raises questions about how much autonomy those interested in participating in research should be allowed and whose interpretation of risk/benefit analysis—the IRB member's, the researcher's, or the potential participant's—should take precedence.

Concerns about how groups are characterized as "vulnerable" in the report have also been voiced. Some worry that the term is not explicitly defined (Rogers and Lange 2013; Levine et al. 2004) and that the reasons for particular groups' vulnerabilities are not explored (King 2005; Luna 2009), leading to a widespread assumption that all individuals in a vulnerable group lack autonomy (Rhodes 2010). Others have suggested that identifying groups or individuals as vulnerable can trigger paternalism and stereotyping (Rogers and Lange 2013; Luna 2009), while others note the failure of the report to spell out different guidelines for groups facing consent-based and fairness-based vulnerability (Nickel 2006).

Ethical Principles and Applications

A final criticism concerns the relationship between ethical principles and their applications. Belmont advocates for a deductive relationship, in that each principle is matched with one aspect of the research process: Respect for persons applies to informed consent, beneficence to risk/benefit analysis, and justice to subject selection.²

2. It should be noted that there is some disagreement among members of the commission about whether the principles were meant to have a deductive relationship with the applications (Beauchamp 2008; Levine 2005). The report itself, however, is quite explicit about the relationship: "Just as the principle of respect for persons finds expression in the requirements for consent, and the principle of beneficence in risk/benefit assessment, the principle of justice gives rise to moral requirements that there be fair procedures and outcomes in the selection of research subjects." The report also suggests that "other principles may also be relevant", but maintains that these three are comprehensive in themselves (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research 1979).

Many criticisms of the report stem from this deductive relationship, as they unpack the implications of putting forward principles with such a narrow scope. For example, several authors suggest that the principle of respect for persons is often taken to represent autonomy, consent, and individual choice, when in fact it ought to have (and some say it used to have) a much broader meaning—concern for a patient and respect for the dignity of each individual, regardless of his or her capacity for autonomous choice (Lebacqz 2005; Faden, Mastroianni, and Kahn 2005; Cassell 2005). A narrow focus on autonomy ignores the way in which people are embedded in relationships, and so does not capture the lived experience of many potential participants (Lebacqz 2005). Similarly, the report's concept of beneficence has been criticized for failing to encompass the related principle of nonmaleficence (Beauchamp 2005; Lebacqz 2005), or for being ambiguous in terms of whether it includes non-maleficence (Veatch 2005). It has also been observed that beneficence has shifted away from meaning the good of the whole patient over time toward a much narrower meaning, measured by the good done in the short run to some dimension of the patient (Cassell 2005). Likewise, the report's discussion of the principle of justice has received heated criticism for its lack of breadth and its narrow focus on subject selection (King 2005). The principle, it has been argued, fails to take into account social justice, including historical patterns of oppression and domination (Shore 2006; Sherwin 1992), and compensatory justice for those injured in research and for groups excluded from the research process (Vanderpool 2001; Lebacqz 2005; King 2005). Justice Belmont-style also does not address procedural justice such as involving vulnerable groups in the process of establishing research agendas (Sherwin 2005; King 2005), or key aspects of distributive justice, especially how the benefits of research will be made available to participants and to underserved populations (Vanderpool 2001).

Questions about how to apply the report's ethical principles are especially pertinent to discussions of IRBs. In an empirical exploration of how IRBs function, investigators found a significant portion of committee discussion revolved around the process of informed consent, while issues related to confidentiality, subject selection, risk/benefit analysis, adverse event reporting, and the minimization of risks were overlooked (Lidz et al. 2012).³ Transcripts from meetings of the National Commission show its concern for mechanisms for the evaluation and monitoring of IRB performance, yet its suggestions include only process measures (site visits, audits of IRB records, and the like), and they were never addressed or published (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research 1978, March 10–11).

3. See also Klitzman (2015).

PART III: MOVING BELMONT INTO THE 21ST CENTURY—IS A TOTAL OVERHAUL NEEDED?

The Line Between Research and Practice Is Insufficient

Defining research and practice as interventions aimed at knowledge and interventions aimed at well-being is incompatible with Belmont's own principles of respecting persons, maximizing beneficence, and ensuring justice. The boundaries around what requires oversight should be defined pragmatically, so that knowledge production and benefits to all communities are maximized and harms to participants are minimized. Oversight should be required for any research or intervention involving novel, significant risks that are not part of the standard of care. This would encourage low-risk data collection during routine practice and ensure that innovative therapies are regulated and their findings disseminated. Investigators ought to be permitted to engage in low-risk or minimally risky research with minimal oversight, while regulatory mechanisms for experiments with significant risks should be kept in place. Regulation ought to be proportionate to novelty and level of risk, rather than derived from intent.

Unique Harms to Communities Remain Unacknowledged

The Belmont Report's failure to consider the potential for harms to communities should be taken into account in the design of any research protocol. This oversight could be overcome if harms to communities were explicitly acknowledged, whether as a new principle or as an extension of the principle of respect for persons. Such a principle "would obligate researchers to respect the values and interests of the community in research and, wherever possible, to protect the community from harms" (Weijer and Anderson 2002). These goals can be achieved by conducting research that is more participatory, by taking into account the needs and interests of the community involved throughout the development of a research agenda, in stages of research collection and analysis, and during the dissemination of results (Weijer and Anderson 2002). A move toward this goal was taken in the recent Common Rule updates, which state that the official governing body of American Indian/Alaska Native tribes can establish additional protections and use of a single IRB (Federal Policy for the Protection of Human Subjects 2017). There is no simple way to define a community, but relevant factors would include self-definition, shared health data and outcomes, group-based vulnerabilities, and historical relationships with prior research.

The Report Fails to Address Current Concerns Involving Transparency

In order for regulatory agencies to operate with the principle of beneficence in mind, they must understand, to the fullest extent possible, the risks and benefits involved in a given research protocol. To achieve this goal, each stage of

the research process must strive for greater transparency. By adopting an additional principle of transparency, or making explicit the principle of nonmaleficence, a revised framework could better address issues that arise in the wake of the commodification of biomedicine. Some advances toward this goal are already taking place, such as the Food and Drug Administration (FDA) requirement that details of all clinical trials be reported on ClinicalTrials.gov (Piller 2015). As industry funding continues to outpace government investments in research, transparency will remain an important tool for minimizing conflicts of interest.

A Focus on Protectionism Is Incompatible With Today's Emphasis on Participation

The Belmont Report's emphasis on researchers' duty to protect participants is admirable and necessary. Yet this duty should be augmented by a duty to include individuals from excluded and vulnerable groups in the research process. Inclusion should be understood to mean including those who have been left out not only as participants but as research partners who can help shape the research goals and protocols. This broader understanding of inclusion is likely to be more effective at reducing the inequalities in both knowledge and opportunity to obtain benefits that exist between those who have been overly protected and those who have not. This move away from protectionism and toward participation, from subjects to partners, is supported by both the principle of beneficence, since it may lead to greater positive health outcomes, and the principle of respect for persons, since many individuals desire to be more involved in research. This also aligns with the recently released Common Rule updates, which now exclude pregnant women and those with physical disabilities from those "potentially vulnerable due to undue influence" (Federal Policy for the Protection of Human Subjects 2017), and the revised Council for the International Organizations of Medical Sciences (CIOMS) International Ethical Guidelines for Health-related Research Involving Humans, which recognize that a judgment of vulnerability "requires empirical evidence to document the need for special protections" (CIOMS 1993). Of course, care must still be taken to ensure that each stage of the research process is not exploitative. Especially important is that participants receive reasonable compensation for harms and injuries they might sustain within an experiment, a problem that remains unsolved to this day.

The Deductive Relationship Between Single Ethical Principles and Applications Espoused by the Report Is Too Limited

Although the Belmont Report suggested that the relationship between individual principles and applications be understood as a deductive one, this unnecessarily

limits the principles to narrow aspects of the research process; the principles instead ought to be understood as applying broadly to all aspects of research involving human subjects. Each principle should be kept in mind by investigators, oversight committees, and regulators throughout each stage of research design, development, implementation, and distribution. Respect for persons should no longer be thought of as a principle that is captured merely by the process of informed consent, but as one that guides the entire research process. In practice, this would involve determining eligibility criteria with this principle in mind, so that individuals are not excluded for trivial reasons (e.g., extra time required for a translator) and those deemed eligible would be likely to benefit from the results (e.g., individuals living below the poverty level are less likely to benefit from an investigation of a novel device not covered by Medicaid). This would also imply that specific informed consent can sometimes be forgone in favor of “broad consent” (Federal Policy for the Protection of Human Subjects 2017; Council for International Organizations of Medical Sciences 2002).⁴ Similarly, follow-up should be conducted with the principle of respect for persons in mind; test results should be given directly to a participant’s primary care physician when requested, and someone from the study team should be available to answer questions for a reasonable period of time once data collection is complete.

CONCLUSION

The Belmont Report was an impressive and inspiring response to the ethical issues of its day. Since its release, the field of research involving human subjects has developed in complex and unexpected ways, challenging the report’s ethical framework to respond not only to the fears related to research abuses that it stemmed from, but also to the increasing commodification of biomedicine, the exclusion of many groups from research, the globalization of research, the desires of many to have access to experimental drugs, the lack of generalizability and reproducibility of many research findings, and the unique harms and histories that communities have experienced as a result of research. While these challenges are likely to continue to shift and expand in the coming years, there are several areas where the report comes up short today. Considering the important role the Belmont Report has played and continues to play in research ethics today (Federal Policy for the Protection of Human Subjects 2017), it is time for a tune-up, if not a complete overhaul. ■

4. In fact, the recently released Common Rule acknowledges that feedback on the previously proposed policy of requiring informed consent for the use of biospecimens was insufficient in that it emphasized “respect for persons with little regard for the principles of beneficence and justice” (Federal policy for the protection of human subjects 2017).

REFERENCES

- Beauchamp, T. L. 2005. The origins and evolution of the Belmont report. In *Belmont revisited: Ethical principles for research with human subjects*, ed. J. F. Childress, E. M. Meslin, and H. T. Shapiro, 12–25. Washington, DC: Georgetown University Press.
- Beauchamp, T. L. 2008. The Belmont report. In *The Oxford textbook of clinical research ethics*, ed. E. J. Emanuel, 149–55. New York, NY: Oxford University Press.
- Beauchamp, T. L., and Y. Saghai. 2012. The historical foundations of the research-practice distinction in bioethics. *Theoretical Medicine and Bioethics* 33 (1):45–56.
- BEECHER, H. 1966. ETHICS AND CLINICAL RESEARCH. *NEW ENGLAND JOURNAL OF MEDICINE* 274 (24):1354.
- Benitez, K., and B. Malin. 2010. Evaluating re-identification risks with respect to the HIPAA privacy rule. *Journal of the American Medical Informatics Association* 17 (2):169–77.
- Cassell, E. J. 2005. The principles of the Belmont report: How have respect for persons, beneficence, and justice been applied in clinical medicine? In *Belmont revisited: Ethical principles for research with human subjects*, ed. J. F. Childress, E. M. Meslin, and H. T. Shapiro, 77–95. Washington, DC: Georgetown University Press.
- Council for International Organizations of Medical Sciences. 1993. International ethical guidelines for biomedical research involving human subjects. http://www.cioms.ch/publications/guidelines/guidelines_nov_2002_blurb.htm.
- Council for International Organizations of Medical Sciences. 2002. International ethical guidelines for biomedical research involving human subjects. *Bulletin of Medical Ethics* Oct. (182):17.
- Dumit, J. 2012. *Drugs for life: How pharmaceutical companies define our health*. Durham, NC: Duke University Press.
- Emanuel, E. J. 2015. Reform of clinical research regulations, finally. *New England Journal of Medicine* 373 (24):2296–99.
- Emanuel, E. J., and C. Weijer. 2005. Protecting communities in research: From a new principle to rational protections. In *Belmont revisited: Ethical principles for research with human subjects*, ed. J. F. Childress, E. M. Meslin, and H. T. Shapiro, 165–83. Washington, DC: Georgetown University Press.
- Faden, R. R., N. E. Kass, S. N. Goodman, P. Pronovost, S. Tunis, and T. L. Beauchamp. 2013. An ethics framework for a learning health care system: A departure from traditional research ethics and clinical ethics. *Hastings Center Reports* 43 (s1):S16–27.
- Faden, R. R., A. C. Mastroianni, and J. P. Kahn. 2005. Beyond Belmont: Trust, openness, and the work of the advisory committee on human radiation experiments. In *Belmont revisited: Ethical principles for research with human subjects*, ed. J. F. Childress, E. M. Meslin, and H. T. Shapiro, 41–54. Washington, DC: Georgetown University Press.
- Federal policy for the protection of human subjects. 2017. *Federal Register* 82 (12):7150, 7158, 7204.
- Gelinas, L., R. Pierce, S. Winkler, and B. Bierer. 2017. Using social media as a research recruitment tool: Ethical issues and recommendations. *American Journal of Bioethics* 17 (3):3–14.
- Jonsen, A. R. 1988. *The abuse of casuistry*. Berkeley, CA: University of California Press.

- Jonsen, A. R. 2005. On the origins and future of the Belmont Report. In *Belmont revisited: Ethical principles for research with human subjects*, ed. J. F. Childress, E. M. Meslin, and H. T. Shapiro, 3–11. Washington, DC: Georgetown University Press.
- Kelly, P., S. J. Marshall, H. Badland, et al. 2013. An ethical framework for automated, wearable cameras in health behavior research. *American Journal of Preventive Medicine* 44 (3):314–19.
- King, P. A. 2005. Justice beyond Belmont. In *Belmont revisited: Ethical principles for research with human subjects*, ed. J. F. Childress, E. M. Meslin, and H. T. Shapiro, 136–47. Washington, DC: Georgetown University Press.
- Klitzman, R. 2015. *The ethics police?: The struggle to make human research safe*. New York, NY: Oxford University Press.
- Lebacqz, K. 2005. We sure are older but are we wiser? In *Belmont revisited: Ethical principles for research with human subjects*, ed. J. F. Childress, E. M. Meslin, and H. T. Shapiro, 99–110. Washington, DC: Georgetown University Press.
- Levine, C., R. Faden, C. Grady, D. Hammerschmidt, L. Eckenwiler, and J. Sugarman. 2004. The limitations of "vulnerability" as a protection for human research participants. *American Journal of Bioethics* 4 (3):44–9.
- Levine, R. J. 1982. Validity of consent procedures in technologically developing countries. In *Human experimentation and medical ethics: Proceedings of the 15th CIOMS Round Table Conference*, ed. Z. Bankowski and R. J. Levine, 16–30. Geneva, Switzerland: Council for International Organizations of Medical Sciences.
- Levine, R. J. 2005. The National Commission's ethical principles with special attention to beneficence. In *Belmont revisited: Ethical principles for research with human subjects*, ed. J. F. Childress, E. M. Meslin, and H. T. Shapiro, 126–35. Washington, DC: Georgetown University Press.
- Lidz, C., P. Appelbaum, R. Arnold, et al. 2012. IRBs: How closely do they follow the Common Rule. *Academic Medicine* 87:1–6.
- Luna, F. 2009. Elucidating the concept of vulnerability: Layers not labels. *International Journal of Feminist Approaches to Bioethics* 2 (1):121–39.
- McCallum, J. M., D. M. Arekere, B. L. Green, R. V. Katz, and B. M. Rivers. 2006. Awareness and knowledge of the U.S. public health service syphilis study at Tuskegee: Implications for biomedical research. *Journal of Health Care for the Poor and Underserved* 17 (4):716–33.
- Metcalf, J., and K. Crawford. 2016. Where are human subjects in big data research? *The emerging ethics divide*. Big Data and Society, May 14.
- National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research. 1978, March 10–11. Meeting Transcript, Tab 2, Box 30. Georgetown University Archives, Washington, DC.
- National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research. 1978, January 13. Meeting Transcript, Tab 2, Box 33. Georgetown University Archives, Washington, DC.
- National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research. 1979. The Belmont report: Ethical principles and guidelines for the protection of human subjects of research. Available at: <http://www.hhs.gov/ohrp/regulations-and-policy/belmont-report>
- Nickel, P. J. 2006. Vulnerable populations in research: The case of the seriously ill. *Theoretical Medicine and Bioethics* 27 (3):245–64.
- The Nuremberg Code. 1949. In *Trials of war criminals before the Nuremberg Military Tribunals under Control Council Law No. 10*, vol. 2, 181–82. Washington, DC: U.S. Government Printing Office.
- Piller, C. 2015. Failure to investigate: A STAT investigation of clinical trial reporting. *STAT News*, December 13. Available at: <https://www.statnews.com/2015/12/13/clinical-trials-investigation/>
- Rhodes, R. 2010. Rethinking Research Ethics. *American Journal of Bioethics* 5(1): 7–28.
- Rogers, W., and M. M. Lange. 2013. Rethinking the vulnerability of minority populations in research. *American Journal of Public Health* 103 (12):2141–46.
- Sherwin, S. 1992. *No longer patient: Feminist ethics and health care*, 72–79. Philadelphia, PA: Temple University Press.
- Sherwin, S. 2005. Belmont revisited through a feminist lens. In *Belmont revisited: Ethical principles for research with human subjects*, ed. J. F. Childress, E. M. Meslin, and H. T. Shapiro, 148–64. Washington, DC: Georgetown University Press.
- Shore, N. 2006. Re-conceptualizing the Belmont report: A community-based participatory research perspective. *Journal of Community Practice* 14 (4):5–26.
- Taylor, P. L. 2010. Overseeing innovative therapy without mistaking it for research: A function-based model based on old truths, new capacities, and lessons from stem cells. *Journal of Law, Medicine & Ethics* 38 (2):286–302.
- Vanderpool, H. Y. 2001. Unfulfilled promise: How the Belmont report can amend the Code of Federal Regulations Title 45 Part 46—Protection of human subjects. *National Bioethics Advisory Commission [NBAC] Ethical and Policy Issues in Research Involving Human Participants Volume II: Commissioned Papers*. Rockville, MD: National Bioethics Advisory Commission.
- Veatch, R. M. 2005. Ranking, balancing, or simultaneity: Resolving conflicts among the belmont principles. In *Belmont revisited: Ethical principles for research with human subjects*, ed. J. F. Childress, E. M. Meslin, and H. T. Shapiro, 184–204. Washington, DC: Georgetown University Press.
- Weijer, C. 1999. Protecting communities in research: philosophical and pragmatic challenges. *Cambridge Quarterly of Healthcare Ethics* 8 (4):501–13.
- Weijer, C., and J. A. Anderson. 2002. A critical appraisal of protections for aboriginal communities in biomedical research. *Jurimetrics* 42 (2):187–98.
- WMA General Assembly. 1996. *WMA Declaration of Helsinki—Ethical principles for medical research involving human subjects*. Somerset West, South Africa: World Medical Association.